

1. **Experiment:** sgRNA Primer Design for dCas9-AID*Δ Targeted Mutagenesis
2. **Time:** 2026.04.02-2026.04.19
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4. **Method:**

CRISPR-Cas9, a third-gen gene-editing tool, excels at precise gene modification. It works by using guide RNA and the Cas9 protein to spot and cut the target genomic DNA in cells, treating it like foreign or viral DNA. The design and features of single-guide RNA are vital for successful gene knockout and function changes. The core principles for sgRNA primer design include compliance with the protospacer adjacent motif (PAM) requirement, appropriate length, high specificity, GC content, avoidance of secondary structures, transcriptional compatibility, and target site positioning. Priority should be given to achieving both high specificity and robust cleavage efficiency. The dCas9-AID*Δ system uses an sgRNA to recruit catalytically inactive Cas9 fused to dCas9-AID*Δ to a selected DNA region, enabling targeted nucleotide diversification without Cas9-mediated double-strand cleavage.

(1) Core Components:

- ① National Center for Biotechnology Information (NCBI) website: As an authoritative sequence resource for CRISPR gRNA design, NCBI supplies standard reference genomes and gene annotations to support off-target evaluation, homology arm design and post-editing verification.
- ② CRISPOR: It can quantitatively predict target editing efficiency and genome-wide off-target risks, providing core evidence for screening high-activity and low-risk optimal gRNAs.
- ③ SnapGene: This software enables gRNA target localization, simulation of expression vector construction and sequence annotation verification, and offers sequence-level support for vector construction and sequence validation in CRISPR editing.

(2) The procedure for sgRNA design:

- ① Target Gene Information Acquisition
 - Reference Genome Confirmation: Select Homo sapiens NCBI GCF_000001405.40 (GRCh38.p14) as the standard human reference genome for sequence alignment and off-target assessment.
 - Genomic Data Retrieval: Extract target gene annotations and corresponding genomic sequences from the NCBI database to define the boundary of the editing region.
- ② Target Region Localization and Verification
 - Sequence Import and Visualization: Import the target genomic sequence into SnapGene to display the full sequence and gene structure features.
 - Editing Region Validation: Identify and confirm the precise target editing region (e.g., functional coding exons) on the sequence map to ensure the accuracy of subsequent design.
- ③ Candidate sgRNA Generation with CRISPOR
 - Parameter Setting and Submission: Submit the validated target sequence to the CRISPOR web tool, and select the matched PAM motif (20 nt-NGG for SpCas9) and corresponding reference genome.
 - Candidate Output: Generate all eligible candidate sgRNA sequences with attached quantitative evaluation metrics.
- ④ Candidate sgRNA Screening and Prioritization
 - Multi-Dimensional Evaluation: Assess each candidate based on on-target editing efficiency (Doench '16 score), genome-wide specificity (MIT and CFD scores), and off-target site distribution.
 - Optimal Candidate Selection: Filter out sgRNAs with high on-target activity and low off-target risk as final candidates for subsequent wet-lab validation.

5. **Result:**

Following the confirmation of the target editing region, CRISPOR was employed to design candidate sgRNAs against the human reference genome. For the dCas9-AID*Δ system, candidates were first filtered based on

whether the dCas9-AID*Δ mutational window (-20 bp upstream of the PAM) covered the target nucleotides. We generated a comprehensive scoring table for each candidate, including MIT score, CFD score, stratified off-target site counts and on-target efficiency, and screened out high-activity sgRNAs with low off-target risk for subsequent experiments.

Table 1. The Sequence of sgRNA.

sgRNA	Sequence	MIT score	CFD score	potential off-target sites
A2-sgRNA-77 forw	AGAATTCTGAACCAAAGTACA	53	81	203
A2-sgRNA-116 forw	GGCTAAGAAGAACCTTCCCG	83	91	84
A2-sgRNA-119 forw	TAAGAAGAACCTTCCCGAGG	84	87	97
A2-sgRNA-122 forw	GAAGAACCTTCCCGAGGAGG	66	79	190
A2-sgRNA-108 rev	TCAGCACCTCCTCCTCGGGA	70	79	232
A3-sgRNA-74 forw	AGAGGCTATCAAACGTACG	91	95	71
A3-sgRNA-77 forw	GGCTATCAAACGTACGAGG	97	97	30
A3-sgRNA-89 forw	GTACGAGGAGGCATTGAAGA	69	87	158
A3-sgRNA-116 forw	ATCTAAGAAGCTGTACTACA	79	86	139
A3-sgRNA-147 forw	GCCTTCGCATATTACGAAC	98	99	13
B2-sgRNA-72 forw	ACCGTAACTGTCTCTTGAC	85	95	63
B2-sgRNA-95 forw	ACCCGGCGAGTACACGTGGG	89	96	32
B2-sgRNA-53 rev	TCCGGTGCAAGAGACAGTTA	80	94	67
B2-sgRNA-78 forw	ACTGTCTCTTGACCCGGACC	89	96	64
B2-sgRNA-91 forw	CCGGACCCGGCGAGTACACG	96	99	26
B2-sgRNA-91 rev	CCACGTGTACTCGCCGGGTC	97	99	16
B2-sgRNA-92 forw	CGGACCCGGCGAGTACACGT	97	99	18
B3-sgRNA-101 forw	GACTCCGAGCGATCTCGAGT	99	100	7
B3-sgRNA-104 forw	TCCGAGCGATCTCGAGTTGG	96	99	16
B3-sgRNA-85 rev	ACCTCCAACCTCGAGATCGCT	97	98	25
B3-sgRNA-117 rev	CGATTGCCAGCACTCCCTCG	92	96	44
B3-sgRNA-118 rev	GCGATTGCCAGCACTCCCTC	78	93	82
C2-sgRNA-138 forw	CACGCCGCCACTCATCCGAA	94	98	33
C2-sgRNA-122 rev	GGCGCCATTCGGATGAGTGG	86	92	67
C2-sgRNA-125 rev	GTTGGCGCCATTCGGATGAG	97	91	44
C2-sgRNA-149 forw	TCATCCGAATGGCGCCAAC	97	98	21
C2-sgRNA-133 rev	GCGGCCAAGTTGGCGCCATT	95	98	36
C3-sgRNA-53 forw	GGACCCGGTTCGAAATTAGGC	98	96	25
C3-sgRNA-61 forw	TCGAAATTAGGCTGGACGAC	95	97	37
C3-sgRNA-104 forw	CGATCGCGTGGAGGGCGACG	92	96	37
C3-sgRNA-134 rev	GGTCACCCGGTGCACGCCCG	93	97	42

Table 2. The Selected sgRNA.

sgRNA	Sequence	MIT score	CFD score	potential off-target sites
A2-sgRNA-116 forw	GGCTAAGAAGAACCTTCCCG	83	91	84
A3-sgRNA-147 forw	GCCTTCGCATATTACGAACT	98	99	13
B2-sgRNA-92 forw	CGGACCCGGCGAGTACACGT	97	99	18
B3-sgRNA-101 forw	GACTCCGAGCGATCTCGAGT	99	100	7
C2-sgRNA-149 forw	TCATCCGAATGGCGCCAACT	97	98	21
C3-sgRNA-53 forw	GGACCCGGTCGAAATTAGGC	98	96	25

Table 3. Optimal sgRNA candidates and their sequences.

sgRNA	Sequence
A2-sgRNA-116 forw	GGCTAAGAAGAACCTTCCCG
B3-sgRNA-101 forw	GACTCCGAGCGATCTCGAGT
C3-sgRNA-53 forw	GGACCCGGTCGAAATTAGGC

6. AI Use Statement:

AI-assisted sequence design was performed. Computational predictions were distinguished from experimental validation.